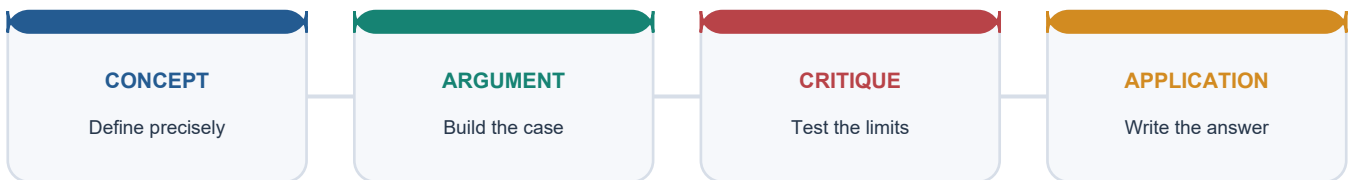


COMPLETE LEARNING SESSION

Logical Reasoning Coding Counting and DI - CSAT Learner-v2 Semantic Successor

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Logical Reasoning Coding Counting and DI - Complete CSAT Learning Session

Identity: csat-06: learner-v2:g1 **Generation date:** 2026-09-06 **Approval:** false **Official syllabus anchor:** Logical reasoning, analytical ability, decision making, problem solving and data interpretation.

Source	SHA-256 at generation
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The canonical Basic owner is taught first. Optional Advanced material is isolated after practice. All official-PYQ references preserve the repository's supplied/provisional key labels; no unavailable wording, official explanation or official key is invented.

BASIC LEARNING SESSION

Twelve-panel ASCII master flow

PANEL 01 - LOGIC VOCABULARY	
Separate implication, converse, contrapositive and equivalence.	
v	
PANEL 02 - PATTERNS AND SERIES	
Test the simplest stable rule and verify every term.	
v	
PANEL 03 - CODING-DECODING	
Infer a rule from all examples, not one coincidence.	
v	
PANEL 04 - DIRECTIONS	
Use coordinates and final displacement.	
v	
PANEL 05 - RELATIONS	
Build a labelled family graph without assuming gender or generation.	
v	
PANEL 06 - SYLLOGISMS	
Use set containment and test existential claims carefully.	
v	
PANEL 07 - ARRANGEMENTS	
Place fixed constraints before flexible ones.	
v	
PANEL 08 - COUNTING	
Apply product, sum, permutation and combination rules with exclusions.	
v	
PANEL 09 - PROBABILITY	
Define equally likely outcomes and conditional sample spaces.	
v	
PANEL 10 - DATA INTERPRETATION	
Read labels, units, bases and totals before calculating.	
v	
PANEL 11 - DECISION SCENARIOS	
Separate facts, constraints, objectives and feasible options.	
v	
PANEL 12 - VERIFICATION	
Enumerate small cases, reverse codes and sanity-check totals.	

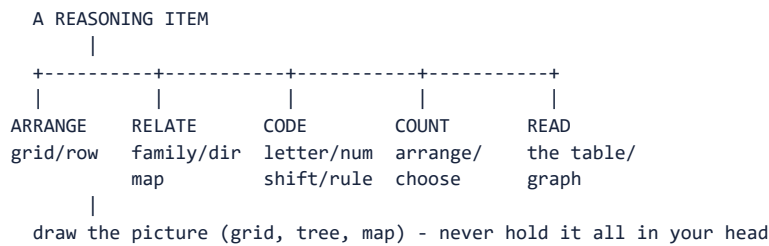
Logical Reasoning, Coding, Counting and Data Interpretation - MUST-DO

Subject: CSAT | **Tier:** Must-Do (foundation) | **Family:** Logical reasoning / decision-making / data interpretation.

Core skill: arrangements, blood relations, directions, coding, series, basic counting/probability, data tables and scenario reasoning. **Boundary:** Topic 07 owns the complete official interpersonal/communication clause; Topic 08 integrates this file into the General Mental Ability map. **Grounded in:** audited UPSC CSAT PYQ sets 2024-2026 (Set A). Drills are **computationally verified**. **FACT:** = rule/fact | **ANALYSIS:** = guidance | $\square$ = original example.

Companion: Advanced Drill.

1. Visual foundation



Core proposition: reasoning items are solved by **externalising** the structure - a grid, a family tree, a direction map - so the constraints do the work.

2. Essential tools

Sub-skill	Method
FACT: Linear/circular arrangement	Draw seats; for a circle, fix one person as an anchor before placing the strongest clue.
FACT: Blood relations	Build a small family tree; resolve "only son/daughter" carefully.
FACT: Directions	Sketch N-E-S-W; each right turn is +90 clockwise.
FACT: Coding-decoding	Find the rule (shift, reverse, position-value) from the given pair, then apply.
FACT: Series	Test differences, ratios, and known sequences (squares, triangular, primes).
FACT: Visual/figure reasoning	Copy the figure; count by size/layer and track invariants (sides, regions, shading, orientation). For cube views, faces visible together are adjacent.
FACT: Conditional/statement logic	Rewrite every "if ... then ..." as $A \rightarrow B$; chain implications and use contrapositives; never the converse.
FACT: Counting	Permutations $P(n, r) = \frac{n!}{(n-r)!}$; combinations $C(n, r) = \frac{n!}{r!(n-r)!}$.
FACT: Probability	$P = \frac{\text{favourable}}{\text{total}}$ for equally likely outcomes; for a sequence, multiply conditional probabilities.
FACT: Data interpretation	Read the title, unit, scale and both axes before computing totals, differences, percentages, shares or weighted values.
FACT: Interpersonal/communication scenario	Apply the dedicated Topic 07 decision engine; this file retains only the reasoning interface.
FACT: Communication matching	Distinguish relationship type, tool purpose, channel, feedback, and barrier from the function described.
FACT: Questionnaire design	Use clear, neutral, single-issue questions; avoid leading, double-barrelled, ambiguous, or intrusive wording.

The provisional 2026 Set-A audit places six explicit interpersonal/communication items at Q72-Q77. Their complete concepts now reside in Topic 07. This remains a year-specific observed block, not evidence that every CSAT paper uses the same count.

MEMORY: Arrangement rule: always start from the clue that **fixes the most**; a single anchored position usually cascades to the full solution.

3. Method

1. Classify the sub-skill and **draw the structure** (row, circle, tree, compass, table).
2. Enter the **strongest constraint** first, then the next, checking consistency each time.
3. For counting/probability, decide **order matters?** (permutation) or **not** (combination) before computing.
4. For DI, answer the **exact** quantity asked (total / change / percent), not a nearby one.
5. For communication scenarios, identify the misunderstanding or conflict, reject coercive or deceptive options, and prefer a clear, fair response with follow-up.

4. Formulas and rules

- FACT: $P(n, r) = n! / (n-r)!$ (arrangements, order matters). *Requires* $0 \leq r \leq n$.
- FACT: $C(n, r) = n! / (r!(n-r)!)$ (selections, order does not matter). *Requires $0 \leq r \leq n$; note $C(n, r) = C(n, n-r)$.*
- FACT: $P(n, r) = C(n, r) \times r!$ - permutations are combinations **times** the orderings of the chosen set.
- FACT: Arrangements of a word with repeats: $\text{total!} / (\text{product of repeat factorials})$.
- FACT: Circular arrangements of n distinct people, where rotations are identical, are $(n-1)!$: fix one person, then arrange the rest. Do **not** divide by 2 unless mirror images are explicitly identical.
- FACT: For an r -digit numeral using a digit set containing zero, choose the first digit from the non-zero digits; then fill the remaining positions under the repetition rule.
- FACT: $P(\text{not } E) = 1 - P(E)$; for equally likely outcomes, $P = \text{favourable} / \text{total}$. *Every probability lies in $[0, 1]$; an answer outside that range is a signal you counted the sample space wrongly.*
- FACT: Sequential probability uses $P(A \text{ and } B) = P(A) \times P(B | A)$. Without replacement, the second fraction changes; multiplication itself remains valid because it uses the **conditional** fraction.
- FACT: $|A \cup B| = |A| + |B| - |A \cap B|$. For a total of N , overlap lies between $\max(0, |A| + |B| - N)$ and $\min(|A|, |B|)$.
- FACT: In a worst-case "minimum draws to ensure" question, first take as many as possible without the target, then add one. For a grid/figure count, list each size exactly once rather than scanning visually.
- FACT: Handshakes or pairwise segments among n objects/points $= C(n, 2)$; these are distinct infinite lines only when no three points are collinear.

4.1 Probability vs possibility

Term	Meaning	Typical stem
FACT: Possible	At least one favourable case exists ; the probability is > 0 .	"Can X happen?", "Is it possible that ...?"
FACT: Certain	Every case is favourable; the probability is 1.	"Must X be true?"
FACT: Probability	The measured fraction favourable/total.	"What is the probability ...?"

- ANALYSIS: "Possible" and "probable" are **not** interchangeable: an event of probability $1/1000$ is fully

possible and highly improbable. In statement-based items, "it is not necessary that Y is correct" means **Y can fail in some consistent case** - one counter-case is enough, no probability needed.

4.2 Conditional statements: implication vs equivalence

```
A -> B  ("if A then B")
|
+-- ALSO TRUE : not B -> not A      (contrapositive - always valid)
+-- NOT GIVEN : B -> A              (converse - a classic trap)
+-- NOT GIVEN : not A -> not B      (inverse - equally invalid)

A <-> B  ("A if and only if B") = BOTH A -> B and B -> A
```

Relation	Symbol	What you may deduce
FACT: Implication	A $\rightarrow$ B	A is sufficient for B; B is necessary for A; use the contrapositive freely.
FACT: Equivalence	A $\leftrightarrow$ B	Both directions; A and B are interchangeable.

- ANALYSIS: **Transitivity is valid for implications:** from "every red is blue", "every blue is green" and "every green is yellow" it follows that every red is green **and** every red is yellow. The audited 2026 Set-A paper contains exactly this chain, and it also contains a four-statement conditional puzzle whose two claims are settled purely by chaining contrapositives.

- ANALYSIS: **Converse is never valid from a one-way statement:** "every red is yellow" does **not** give "every yellow is red".

5. Original solved examples

□ Example A (series - letters)

A, C, F, J, ? Positions 1, 3, 6, 10, 15 (gaps +2, +3, +4, +5; triangular numbers). Next = 15 $\rightarrow$ **O**. (Verified sequence 1,3,6,10,15.)

□ Example B (blood relation)

A male speaker says, "She is the daughter of my paternal grandfather's only son." The paternal grandfather's only son is the speaker's father; the named woman is therefore the speaker's **sister**. **Answer: sister.**

□ Example C (counting)

How many 3-digit numbers with distinct digits can be formed from {1, 2, 3, 4}? $P(4, 3) = 4 \times 3 \times 2 = 24$. (Verified.)

□ Example D (probability)

A bag has 3 red and 2 blue balls; one is drawn. $P(\text{red}) = 3/(3+2) = 3/5$. (Verified.)

□ Example E (data interpretation)

Read this table:

Month	Product X	Product Y
Jan	200	150
Feb	250	180
Mar	300	120

- Total X = 200 + 250 + 300 = **750**; total Y = 150 + 180 + 120 = **450**.
- Percentage rise in X from Jan to Mar = $(300 - 200)/200 = 50\%$. (Verified.)

□ Example F (communication - original)

A colleague follows an outdated procedure after receiving two conflicting emails. What should a team lead do first? Clarify the current procedure privately, explain which instruction controls, ask the colleague to restate it, and circulate one corrected instruction to the team. This addresses the information failure without blame.

□ Example G (implication chain - original)

Given. Every scholarship holder is a hosteller. Every hosteller is a registered student.

Which follow? (i) Every scholarship holder is a registered student. - **Follows** (transitivity of $\rightarrow$). (ii) Every registered student is a hosteller. - **Does not follow** (that is the converse). (iii) Someone who is not a registered student is not a scholarship holder. - **Follows** (that is the contrapositive of (i)).

Answer: (i) and (iii) only. (Verified: the only valid moves from a one-way implication are chaining and contraposition.)

□ Example H (circular arrangement)

Five friends sit around a round table. In how many distinct seatings can they sit if rotations are the same? Fix A at the top to remove rotational duplicates. Arrange B, C, D and E in the remaining seats: $4! = 24$. A reflection is a different seating unless the stem says clockwise and anticlockwise arrangements are identical. (Verified.)

□ Example I (coding verification)

If CAT becomes DBU and DOG becomes EPH, code SUN. Each corresponding letter shifts +1: C $\rightarrow$ D, A $\rightarrow$ B, T $\rightarrow$ U and D $\rightarrow$ E, O $\rightarrow$ P, G $\rightarrow$ H; the second pair verifies that the rule is not guessed from one letter. Hence S $\rightarrow$ T, U $\rightarrow$ V, N $\rightarrow$ O: **TVO**.

□ Example J (Venn/set range)

In a group of 50, 28 study A and 24 study B. The number studying both can be from $\max(0, 28+24-50) = 2$ to $\min(28, 24) = 24$. Thus **2 to 24** are possible; choosing a single overlap without further data is unjustified. (Verified.)

□ Example K (constrained number with zero)

How many **five-digit** numbers can be made from 0, 1, 2, 3 and 4 without repetition? The first place cannot be zero: 4 choices. The remaining four digits can be arranged in $4!$ ways. Total: $4 \times 4! = 96$. (Verified.)

□ Example L (conditional probability without replacement)

A bag has 3 red and 2 blue balls. Given that the first draw is red and it is not replaced, the probability that the second is red is $2/4 = 1/2$. The probability of two reds in order is $(3/5) \times (2/4) = 3/10$: fractions do multiply, but the second is the **conditional** fraction.

□ Example M (table, pie and dual-axis DI)

Quarter	Output (thousand units)	Revenue (Rs lakh)	Product P share of revenue
Q1	40	80	25%
Q2	50	90	30%

- **Bar/table reading:** output rose by 10 thousand, not 10%.

- **Dual-axis reading:** revenue per thousand units in Q2 is $90/50 = 1.8$ lakh, while Q1 is 2; output rose but revenue per unit fell.

- **Pie/share reading:** P's Q2 revenue is 30% of 90 = 27 lakh; its share rose by ****5 percentage points****, or $5/25 = 20\%$ relatively. Name the requested measure before choosing an option.

□ Example N (visual/cube and worst-case execution)

Cube views. A labelled cube shows faces (1, 2, 3) at one corner and (1, 4, 5) at another. Faces 2, 3, 4 and 5 are each adjacent to 1, so the remaining face **6 is opposite 1**. Do not infer an opposite pair from a single view.

Grid count. A 3×3 square grid has 9 unit squares, 4 squares of side 2, and 1 of side 3: **14** total. Count by size, not by visual impression.

Worst case. From red and blue balls, how many draws ensure three of one colour? Take 2 red and 2 blue without reaching three, then one more: **5**.

6. Must-Know facts

- FACT: Order matters -> permutation; order does not -> combination; $P(n, r) = C(n, r) \times r!$.
- FACT: Each **right** turn is 90 degrees clockwise; two rights = a U-turn (180).
- FACT: "Only son/daughter" removes other children of the **same stated sex**, not all siblings; map the maternal/paternal side and the speaker's sex before concluding.
- FACT: For a valid syllogism, a conclusion must hold in **every** arrangement, not just one.
- FACT: From $A \rightarrow B$ you may use the **contrapositive**; you may never use the **converse**.
- FACT: "Possible" only needs one favourable case; "necessary/certain" needs every case.
- FACT: In DI, label ****level / change / percentage change / share / percentage-point change / weighted rate****; a graph axis may use a different unit or scale from the other axis.
- FACT: In circular seating, fix one anchor; in a digit-number count, forbid leading zero before using a permutation.
- FACT: Venn overlap has lower and upper bounds; do not invent an intersection from two set sizes.
- FACT: In interpersonal items, prefer listening, clarity, fairness, legality, proportionality, and confirmation over humiliation, threats, concealment, or impulsive escalation.

7. Common traps

- WRONG: Using permutations when order does not matter (or vice versa). -> Decide first.
- WRONG: Reading "some A are B" as "all A are B". -> "Some" is existential, not universal.
- WRONG: Reading $A \rightarrow B$ as $B \rightarrow A$. -> Only the contrapositive is free.
- WRONG: Treating "possible" as "probable", or "not necessary" as "false". -> "Not necessary" only means **at least one** consistent case makes it fail.
- WRONG: In directions, treating a left turn as +90 clockwise. -> Left is anticlockwise.
- WRONG: In DI, computing a total when a **change** or **percentage** was asked. -> Re-read the stem.
- WRONG: Treating a circular rotation as a new arrangement. -> Fix one anchor before counting.
- WRONG: Allowing zero in the first place of a multi-digit number. -> Choose the first digit separately.
- WRONG: Multiplying unchanged fractions without replacement. -> Multiply sequential **conditional** fractions, or use combinations for an unordered draw.
- WRONG: Reading a graph without its axis unit/scale or treating points as percent change. -> Label both axes and the base before computing.
- WRONG: Declaring cube faces opposite from one view or double-counting a grid figure. -> Use all views and count by size/layer.
- WRONG: Accepting a syllogism conclusion true in one diagram only. -> It must hold in **all**.
- WRONG: Treating a communication problem as a character flaw before clarifying facts. -> Diagnose the information/process failure first.
- WRONG: Treating every communication failure as conflict. -> First classify the barrier: physical, semantic/language, psychological, organisational, or technological.

8. Quick checks

- FACT: Can you decide permutation vs combination from the stem in one read?
- FACT: Can you sketch a family tree from "only son/daughter" clues?
- FACT: Can you write the contrapositive of a given "if ... then ..." instantly?
- FACT: Can you convert a table row into a percentage change quickly?

9. Mini-drill (with answers and explanations)

1. Next term: 5, 11, 23, 47, ...
2. In how many ways can the letters of "LEVEL" be arranged?
3. A man walks 3 km North, turns right and walks 4 km, turns right and walks 3 km. How far is he from the start, and in which direction?
4. Ten people each shake hands once with every other. How many handshakes?
5. Using the Section 5 table, what is the total sales of Product Y over the three months?
6. A bag holds 3 red and 2 blue balls. Two are drawn together at random. Is it **possible** to draw two blue balls, and what is the **probability** of drawing two red ones?
7. Given "if a file is approved, then it is signed" and "this file is not signed", what follows about approval?
8. Six people sit around a round table. How many distinct arrangements are there if rotations are the same?
9. How many four-digit numbers can be formed from 0, 1, 2 and 3 without repetition?
10. Of 60 students, 35 take Hindi and 28 take Tamil. What is the least possible number taking both?
11. A bag has 4 white and 6 black balls. Two are drawn in sequence without replacement. Find $P(\text{white then white})$.
12. A pie slice rises from 20% to 25%. State the percentage-point change and the relative percent change.
13. How many squares are present in a 2×2 square grid?
14. From three colours of balls, how many draws ensure two balls of one colour?

Answers.

1. **95.** Each term is $x^2 + 1$: $47 \times 2 + 1 = 95$. (Verified.)
2. **30.** $5! / (2! \times 2!) = 120 / 4 = 30$ (E and L repeat twice). (Verified.)
3. **4 km, due East.** The two 3 km legs (N then S) cancel; the 4 km East leg remains. (Verified.)
4. **45.** $C(10, 2) = 45$. (Verified.)
5. **450.** $150 + 180 + 120 = 450$. (Verified.)
6. **Yes, possible;** $P(\text{two red}) = 3/10$. Two blue is possible - exactly one such pair exists, so its probability is $1/C(5, 2) = 1/10$, small but non-zero. Two red: $C(3, 2)/C(5, 2) = 3/10$. (Verified: $C(5, 2) = 10$.) ANALYSIS: "Possible" was a yes/no question; "probability" was a number.
7. **The file was not approved.** That is the **contrapositive** of "approved $\rightarrow$ signed", which is always valid. It does **not** follow that a signed file was approved - that would be the converse. (Verified as a logical form.)
8. **120.** Fix one person and arrange the other five: $5! = 120$.
9. **18.** First digit: 3 non-zero choices; then $3!$ arrangements: 3×6 .
10. **3.** The lower overlap bound is $35 + 28 - 60 = 3$.
11. **2/15.** $P(W \blacksquare) \times P(W \blacksquare | W \blacksquare) = (4/10) \times (3/9)$.
12. **+5 percentage points; +25% relative change.** The second uses $5/20$, not $5/25$.

13. **5.** Four unit squares plus one large square.
14. **4.** In the worst case, take one of each colour first, then one more.

10. Timed transfer, diagnosis and retry gate

Attempt items 8-12 plus one seating/coding/DI question of your choice in **nine minutes**. For each miss, diagnose before checking the answer:

Symptom	Code	Repair	
Rotation or reflection double-counted	C/A	Anchor one seat; read whether reflections are distinct.	
Leading zero or repetition rule lost	R	Write allowed first-place digits before any factorial.	
One Venn intersection assumed	C	Write lower and upper overlap bounds.	
Without-replacement fraction unchanged	A	Write 'P(B	A)' explicitly.
Graph/share measure confused	R/X	Label level, share, points, and relative change before arithmetic.	

Retry gate: after a 20-minute gap, score **6/6 in nine minutes** with no C/R error. Otherwise redo the matching worked example and retry fresh values. The Advanced companion is optional speed depth, not a prerequisite for this gate.

11. Study links

- FACT: Optional Advanced companion -

multi-constraint seating, harder syllogisms, and speed depth. Core above independently covers the required arrangement, Venn, coding, counting, probability, and DI mechanisms.

- FACT: Algebra, Inequalities and Data Sufficiency - decision-making shares the sufficiency mindset.
- FACT: Number Systems and Number Sense - counting rests on factor and number sense.

2026 PYQ Integration

Status: 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2026.md. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** CSAT
- **Routed question demands:** 21

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	9	Constrained-word permutation count	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	14	Circular-seating arrangement	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	15	Painted-region pattern count	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	23	Weighing-combination count	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	33	Ranking-order logic	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	34	Segment-matching puzzle	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	35	Group-seating constraints	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	36	Statement-implication logic	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	44	Blood-relation tree	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	48	Transitive-colour syllogism	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	51	Odd-item weighing logic	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	56	Direction turns and distance	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	67	Letter-code decoding	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	70	Weighted-score data table	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	71	Statement-relationship logic	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	72	Communication: relationship matching	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	73	Communication: tool-purpose matching	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	74	Communication: conflict scenario	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	75	Communication: barrier-example matching	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	CSAT	76	Communication: questionnaire scenario	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.
2026	CSAT	77	Communication: statements	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (scan headed CS (P) Exam 2026 [Prov. Ans. Key]); family/neutral type per CSAT/00_Question-Audit-Ledger; key is provisional - no answer letter recorded or inferred here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Constrained-word permutation count
- Circular-seating arrangement
- Painted-region pattern count
- Weighing-combination count
- Ranking-order logic
- Segment-matching puzzle
- Group-seating constraints
- Statement-implication logic
- Blood-relation tree
- Transitive-colour syllogism
- Odd-item weighing logic
- Direction turns and distance
- Letter-code decoding
- Weighted-score data table
- Statement-relationship logic
- Communication: relationship matching
- Communication: tool-purpose matching
- Communication: conflict scenario
- Communication: barrier-example matching
- Communication: questionnaire scenario
- Communication: statements

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2024-2025.md. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** CSAT
- **Routed question demands:** 24

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	24	Sequence completion	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	45	Digit-frequency counting	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	48	Alphametic addition	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	49	Constrained-triplet count	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	56	Multi-turn direction sense	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	57	Rotational direction sense	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	58	Conclusion validity	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	CSAT	74	Implication consistency	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	75	Operator substitution	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	77	Alphabetic-number coding	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	78	Letter-rearrangement coding	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2024	CSAT	80	Coded-value constraints	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	8	Month-pattern sequence	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	29	Alphabetic subtraction	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	30	Letter-pattern completion	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	CSAT	38	Coded-operator arrangement	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	49	Suspect-statement logic	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	50	Tournament-score logic	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	55	Family-relation deductions	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	56	Multiplicative-letter code	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	57	Letter-digit substitution	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	58	Constrained-digit arrangement	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.
2025	CSAT	65	Direction-network distance	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	CSAT	66	Numeric-cube code	Objective question; official Set-A key available locally, answer not inferred	Key supplied locally (Set-A scan; recorded as supplied, not certified final); family and neutral type per CSAT/00_Question-Audit-Ledger; answer not recorded here	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- Sequence completion
- Digit-frequency counting
- Alphametic addition
- Constrained-triplet count
- Multi-turn direction sense
- Rotational direction sense
- Conclusion validity
- Implication consistency
- Operator substitution
- Alphabetic-number coding
- Letter-rearrangement coding
- Coded-value constraints
- Month-pattern sequence
- Alphametic subtraction
- Letter-pattern completion
- Coded-operator arrangement
- Suspect-statement logic
- Tournament-score logic
- Family-relation deductions
- Multiplicative-letter code
- Letter-digit substitution
- Constrained-digit arrangement
- Direction-network distance
- Numeric-cube code

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 127

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	1	3D shape counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	4	Polygon diagonal count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	11	Bar-chart data comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	12	Bar-chart value calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	13	Bar-chart ratio reading	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	14	Cube rotation opposite face	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	15	Cube rotation opposite face	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	16	Cube rotation opposite face	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	21	Letter substitution coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	22	Letter-to-number coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	25	Office seating deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	26	Office seating deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	42	Venn-diagram counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	43	Placement logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	44	Placement logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	45	Placement logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	46	Placement statement check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	47	Placement statement check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	48	Placement statement check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	61	Progress graph comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	62	Color arrangement counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	63	Population graph inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	64	Graph statement validity	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	65	Dual-scale graph comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	66	Figural analogy completion	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	67	Graph break-even reading	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	70	Figural rotation series	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	71	Population age-group graph	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	72	Logical relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	73	Policy rate graph inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	74	Table data inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	75	Table assumption validity	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	76	Tax revenue graph inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	10	Cube partition cuts	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	13	Syllogism conclusion check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	14	Grid parallelogram count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	20	Set overlap deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	29	Statement sufficiency check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	31	Venn diagram deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	32	Logical chain elimination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	52	Venn-diagram counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	54	Letter sequence fill	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	CSAT	56	Denomination combination count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	68	Alphabet substitution coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	70	Sequence conditional count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	71	Blood-relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	72	Cube face coloring count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	73	Natural number partition count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	74	Symbol substitution evaluation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	76	Ranking order deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	77	Ranking order deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	78	Ranking order deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	79	Weight inequality deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	80	Weight inequality deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	15	Letter sequence completion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	19	Blood-relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	32	Syllogism deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	33	Syllogism deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	34	Sequence pair counting	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	35	Linear row position count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	36	Age-ordering deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	48	Letter-number coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	49	Letter analogy series	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	52	Table rate comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	60	Denomination subset counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	67	Constrained permutation count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	10	Syllogism conclusion selection	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	15	Syllogism conclusion selection	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	17	Mirror-image consonant count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	19	Digit-sum integer count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	25	Multi-attribute classification	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	27	Matrix trend missing entry	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	28	Dual-sequence missing entry	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	35	Venn set overlap range	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	40	Alphabet reversal position coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	45	Table run-rate comparison	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	46	Fractional set deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	49	Multi-conclusion syllogism	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	56	Letter-substitution coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	64	Chessboard diagonal count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	65	Series blank-fill pattern	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	66	Restricted digit arrangement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	69	Non-consecutive selection count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	74	Pie-chart sector angle	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	10	Digit row arrangement count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	16	Alphabet reversal coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	17	Elimination match counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	18	Odd digit counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	24	Letter arrangement counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	27	Route combination counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	29	Letter series position	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	30	Queue minimum size	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	36	Blood relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	38	Two-statement syllogism	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	39	Three-statement syllogism	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	40	Circular seating arrangement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	55	Constrained PIN enumeration	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	56	Circle inscribed triangle count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	70	Pie chart percentage	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	76	Constrained password count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	77	Grid row arrangement count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	79	Form constraint number count	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	4	Worst-case pair selection	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	5	Score partition counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	6	Derangement possibility check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	18	Interior cube count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	19	Position-constrained permutation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	24	Symbol operation decode	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	30	Collinearity-aware triangle count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	36	Worst-case colour group draw	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	37	Letter-shift cipher decode	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	38	Statement-based age ranking	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	39	Family-relation statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	40	Prime-group odd-one-out	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	48	Circular consecutive selection count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	54	Circular pass-sequence return	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	55	Sequence middle term	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	65	Integer-constraint solution count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	66	Adjacent-colour stripe count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	68	Constrained assignment count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	69	Weighing-coin optimisation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	70	Coded relational chain deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	75	Repeating pattern gap fill	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	77	True-false logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	78	Conditional logic conclusion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	79	Painted cuboid partition count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	80	Reverse alphabetical position count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- 3D shape counting
- Polygon diagonal count
- Bar-chart data comparison
- Bar-chart value calculation
- Bar-chart ratio reading
- Cube rotation opposite face
- Letter substitution coding
- Letter-to-number coding
- Office seating deduction
- Venn-diagram counting
- Placement logic deduction
- Placement statement check
- Progress graph comparison
- Color arrangement counting
- Population graph inference
- Graph statement validity
- Dual-scale graph comparison
- Figural analogy completion
- Graph break-even reading
- Figural rotation series
- Population age-group graph
- Logical relation deduction
- Policy rate graph inference
- Table data inference
- Table assumption validity
- Tax revenue graph inference
- Cube partition cuts
- Syllogism conclusion check
- Grid parallelogram count
- Set overlap deduction
- Statement sufficiency check
- Venn diagram deduction
- Logical chain elimination
- Letter sequence fill
- Denomination combination count
- Alphabet substitution coding
- Sequence conditional count
- Blood-relation deduction

- Cube face coloring count
- Natural number partition count
- Symbol substitution evaluation
- Ranking order deduction
- Weight inequality deduction
- Letter sequence completion
- Syllogism deduction
- Sequence pair counting
- Linear row position count
- Age-ordering deduction
- Letter-number coding
- Letter analogy series
- Table rate comparison
- Denomination subset counting
- Constrained permutation count
- Syllogism conclusion selection
- Mirror-image consonant count
- Digit-sum integer count
- Multi-attribute classification
- Matrix trend missing entry
- Dual-sequence missing entry
- Venn set overlap range
- Alphabet reversal position coding
- Table run-rate comparison
- Fractional set deduction
- Multi-conclusion syllogism
- Letter-substitution coding
- Chessboard diagonal count
- Series blank-fill pattern
- Restricted digit arrangement
- Non-consecutive selection count
- Pie-chart sector angle
- Digit row arrangement count
- Alphabet reversal coding
- Elimination match counting
- Odd digit counting
- Letter arrangement counting
- Route combination counting
- Letter series position
- Queue minimum size
- Blood relation deduction
- Two-statement syllogism
- Three-statement syllogism
- Circular seating arrangement

- Constrained PIN enumeration
- Circle inscribed triangle count
- Pie chart percentage
- Constrained password count
- Grid row arrangement count
- Form constraint number count
- Worst-case pair selection
- Score partition counting
- Derangement possibility check
- Interior cube count
- Position-constrained permutation
- Symbol operation decode
- Collinearity-aware triangle count
- Worst-case colour group draw
- Letter-shift cipher decode
- Statement-based age ranking
- Family-relation statement sufficiency
- Prime-group odd-one-out
- Circular consecutive selection count
- Circular pass-sequence return
- Sequence middle term
- Integer-constraint solution count
- Adjacent-colour stripe count
- Constrained assignment count
- Weighing-coin optimisation
- Coded relational chain deduction
- Repeating pattern gap fill
- True-false logic deduction
- Conditional logic conclusion
- Painted cuboid partition count
- Reverse alphabetical position count

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Semantic-completeness closure - 2026-09-06

Literal syllabus and canonical ownership

- **Literal clause:** Logical reasoning, analytical ability, decision making, problem solving and data interpretation.
- **Canonical scope:** Owns series, analogies, coding-decoding, directions, relations, syllogisms, arrangements, ranking, counting, permutations, combinations, probability, tables/charts and scenario logic.
- **Cross-topic boundary:** Passage inference belongs to Topic 01; algebraic data sufficiency to Topic 05; communication content to Topic 07 even when a matching format is structurally routed here.

Complete learner route

1. **Logic vocabulary:** Separate implication, converse, contrapositive and equivalence.
2. **Patterns and series:** Test the simplest stable rule and verify every term.

3. **Coding-decoding:** Infer a rule from all examples, not one coincidence.
4. **Directions:** Use coordinates and final displacement.
5. **Relations:** Build a labelled family graph without assuming gender or generation.
6. **Syllogisms:** Use set containment and test existential claims carefully.
7. **Arrangements:** Place fixed constraints before flexible ones.
8. **Counting:** Apply product, sum, permutation and combination rules with exclusions.
9. **Probability:** Define equally likely outcomes and conditional sample spaces.
10. **Data interpretation:** Read labels, units, bases and totals before calculating.
11. **Decision scenarios:** Separate facts, constraints, objectives and feasible options.
12. **Verification:** Enumerate small cases, reverse codes and sanity-check totals.

Verification and hostile-query gate

Logical conclusions are model-checked, counting answers are enumerated for small cases, probabilities use explicit sample spaces, and DI preserves denominator and units.

The hostile absence search explicitly tested these families and close-option terms: **syllogism; coding; direction; blood relation; arrangement; permutation; probability; data interpretation**. A shortcut is usable only when its stated condition survives; otherwise return to the first-principles representation. Every worked answer must finish with an independent check: passage support, substitution, enumeration, units, bounds, reverse operation or option elimination.

Difficulty and timed progression

1. Foundation: recognise the family and state the governing definition or relation.
2. Core: solve a direct item with a visible representation and one verification.
3. Advanced: combine two mechanisms, test edge cases and reject close distractors.
4. Timed: use classify -> extract -> represent -> execute -> verify -> decide.
5. Remediation: log the error as concept, application, calculation, reading, passage, time or guess; return to the owning subtopic before retesting.

BASIC MCQS / REMEDIATION

Diagnostic and core set

Q1. Practice variant 1: Complete the series: 2, 4, 8, 14, ?

A. 22 B. 23 C. 21 D. 24

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q2. Practice variant 1: Each letter is shifted forward by 1. How is CAT coded?

A. CAT B. DBU C. ECV D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q3. Practice variant 1: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?

A. East B. South C. North D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q4. Practice variant 1: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q5. Practice variant 1: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q6. Practice variant 1: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Order matters, so use nPr .

Q7. Practice variant 1: A bag has 2 red and 3 blue balls. One is drawn at random. Probability of red?

A. $\frac{3}{5}$ B. $\frac{1}{2}$ C. $\frac{2}{5}$ D. 1

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q8. Practice variant 1: A table lists three values: 20, 30, 50. What is their total?

A. 101 B. 99 C. 102 D. 100

Correct answer: D. Read the same unit and add all rows once.

Q9. Practice variant 2: Complete the series: 3, 5, 9, 15, ?

A. 23 B. 24 C. 22 D. 25

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q10. Practice variant 2: Each letter is shifted forward by 2. How is CAT coded?

A. DBU B. ECV C. CAT D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q11. Practice variant 2: A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?

A. North B. South C. East D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q12. Practice variant 2: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q13. Practice variant 2: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q14. Practice variant 2: How many ordered selections of 2 people can be made from 6 distinct people?

A. 31 B. 30 C. 29 D. 32

Correct answer: B. Order matters, so use nPr.

Q15. Practice variant 2: A bag has 3 red and 3 blue balls. One is drawn at random. Probability of red?

A. 3/6 B. 1 C. 1/2 D. None of these (1)

Correct answer: C. Favourable equally likely outcomes divided by total outcomes.

Q16. Practice variant 2: A table lists three values: 21, 31, 51. What is their total?

A. 104 B. 102 C. 105 D. 103

Correct answer: D. Read the same unit and add all rows once.

Remediation protocol

1. Recompute without options.
2. Name the failed rule or passage phrase.
3. Reject every distractor for a specific reason.
4. Retry with altered numbers, wording or constraints.
5. Advance only after two consecutive correct answers under the time ceiling.

PYQS AND ANSWER PRACTICE

Verified 2024-2026 Set-A demand and key ledger

Year	Q	Neutral verified demand	Key status	Solution architecture
2024	24	Sequence completion	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	45	Digit-frequency counting	B (supplied)	Use place value, cyclicity or a constrained enumeration.
2024	48	Alphametic addition	A (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	49	Constrained-triplet count	D (supplied)	Use the total crossing length.
2024	56	Multi-turn direction sense	C (supplied)	Use coordinates and final displacement.
2024	57	Rotational direction sense	D (supplied)	Use coordinates and final displacement.
2024	58	Conclusion validity	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	74	Implication consistency	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	75	Operator substitution	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	77	Alphabetic-number coding	A (supplied)	Infer one rule that fits every example and reverse-check.
2024	78	Letter-rearrangement coding	D (supplied)	Place hard constraints in a grid before testing options.
2024	80	Coded-value constraints	B (supplied)	Use the total crossing length.
2025	8	Month-pattern sequence	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.

Year	Q	Neutral verified demand	Key status	Solution architecture
2025	29	Alphabetic subtraction	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	30	Letter-pattern completion	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	38	Coded-operator arrangement	B (supplied)	Place hard constraints in a grid before testing options.
2025	49	Suspect-statement logic	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	50	Tournament-score logic	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	55	Family-relation deductions	C (supplied)	Draw a labelled generation graph.
2025	56	Multiplicative-letter code	A (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	57	Letter-digit substitution	D (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	58	Constrained-digit arrangement	B (supplied)	Use place value, cyclicity or a constrained enumeration.
2025	65	Direction-network distance	C (supplied)	Add rates and invert the net rate.
2025	66	Numeric-cube code	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	9	Constrained-word permutation count	D (provisional)	Use the total crossing length.
2026	14	Circular-seating arrangement	A (provisional)	Place hard constraints in a grid before testing options.
2026	15	Painted-region pattern count	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	23	Weighing-combination count	B (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	33	Ranking-order logic	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	34	Segment-matching puzzle	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	35	Group-seating constraints	B (provisional)	Use the total crossing length.
2026	36	Statement-implication logic	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	44	Blood-relation tree	D (provisional)	Draw a labelled generation graph.
2026	48	Transitive-colour syllogism	D (provisional)	Use set containment and reject an illicit converse.
2026	51	Odd-item weighing logic	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	56	Direction turns and distance	D (provisional)	Use coordinates and final displacement.
2026	67	Letter-code decoding	B (provisional)	Infer one rule that fits every example and reverse-check.
2026	70	Weighted-score data table	A (provisional)	Read units and denominators before aggregation.
2026	71	Statement-relationship logic	D (provisional)	Draw a labelled generation graph.
2026	72	Communication: relationship matching	B (provisional)	Draw a labelled generation graph.

Year	Q	Neutral verified demand	Key status	Solution architecture
2026	73	Communication: tool-purpose matching	C (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	74	Communication: conflict scenario	C (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	75	Communication: barrier-example matching	D (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	76	Communication: questionnaire scenario	C (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.
2026	77	Communication: statements	D (provisional)	Diagnose the barrier or purpose, then choose the feasible inclusive response.

The table is a non-verbatim routing audit. It records the locally checked Set-A key letter and a valid solving route, but does not pretend that UPSC publishes model solutions. The separate workbook supplies original solved equivalents for every mechanism.

Timed mixed transfer

Q33. Practice variant 5: Complete the series: 6, 8, 12, 18, ?

A. 26 B. 27 C. 25 D. 28

Correct answer: A. The successive increments are 2, 4, 6, 8.

Q34. Practice variant 5: Each letter is shifted forward by 2. How is CAT coded?

A. DBU B. ECV C. CAT D. None of these (1)

Correct answer: B. Apply the same alphabetic shift to every letter and reverse-check.

Q35. Practice variant 5: A person faces North and turns right 0 time(s), each by 90 degrees. Which direction is faced?

A. East B. South C. North D. West

Correct answer: C. Represent orientation on a four-direction cycle.

Q36. Practice variant 5: Riya is the daughter of Mohan's only son. How is Riya related to Mohan?

A. Daughter B. Sister C. Niece D. Granddaughter

Correct answer: D. Mohan's son is Riya's parent, so Riya is Mohan's granddaughter.

Q37. Practice variant 5: All poets are readers. No reader is silent. Which conclusion follows?

A. No poet is silent B. All silent people are poets C. Some poets are silent D. No conclusion follows

Correct answer: A. Poets are inside readers, and readers do not overlap silent people.

Q38. Practice variant 5: How many ordered selections of 2 people can be made from 5 distinct people?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Order matters, so use nPr .

Tool	Statement	Condition								
FACT: Dice/coin counting	Enumerate the sample space ($6^k, 2^k$); count favourable.	Outcomes must be equally likely.								
FACT: Drawing without replacement	For an unordered whole draw, use $C(n, r)$; for sequential draws, multiply conditional fractions.	Without replacement, the later fraction changes (e.g. $4/10 \times 3/9$); multiplication remains valid because it is conditional.								
FACT: Inclusion-exclusion		$A \cap B$	$=$	$A \cup B$	$+$	B	$-$	$A \cap B$	; for three sets, subtract the three pairwise overlaps and add back the triple.	Use the actual intersection for arbitrary sets. Use lcm only when the sets are multiples of given integers .
FACT: Polygon diagonals	$n(n-3)/2$	Convex polygon.								

3. Harder methods (worked)

□ Method 1 - multi-constraint seating

Five people P, Q, R, S, T sit in seats 1-5 (left to right). Clues: R is in the middle (seat 3); P is at the extreme left; Q is immediately right of P; T is **not** at an extreme end. Then $P=1, Q=2, R=3$; T cannot be seat 5, so $T=4$ and $S=5$.

Unique order: P, Q, R, T, S. (Each clue checks out; the solution is forced.)

□ Method 2 - coding rule

If EARTH is coded GCTVJ (each letter shifted +2), how is MARS coded? $M \rightarrow O, A \rightarrow C, R \rightarrow T, S \rightarrow U \rightarrow$ **OCTU**. (Verified by letter positions $13 \rightarrow 15, 1 \rightarrow 3, 18 \rightarrow 20, 19 \rightarrow 21$.)

□ Method 3 - dice probability

Two fair dice; $P(\text{sum} = 7)$? Favourable pairs: $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6$ of $36 \rightarrow$ **1/6**. (Verified.)

□ Method 4 - conditional chains: solve them backwards

ANALYSIS: A statement-logic item gives several "if ... then ..." links and asks which claims follow. Do **not** try to assign truth values by trial. Instead:

- Write every link as an arrow: $A \rightarrow B$
- Write the contrapositive beside it: $\text{not } B \rightarrow \text{not } A$
- To test "if P then Q", ASSUME P and not Q, then chain to a contradiction.
- To test "it is not necessary that Q", find ONE consistent case with Q false.

Worked example (original). Four claims W, X, Y, Z satisfy: (i) if X is false then Z is false; (ii) if Y is false then W is true; (iii) if W is true then X is false.

- **Does "X true $\rightarrow$ Y true" hold?** Assume X true and Y false. By (ii), W is true. By (iii), X is false - contradicting the assumption. So **Y must be true**: the claim **holds**.

- **Is it possible for Z to be true while Y is false?** (i) contrapositive is "Z true $\rightarrow$ X true", and we just showed X true forces Y true. So **no** - "Z true with Y false" is impossible, i.e. the claim "if Z is true it is *not* necessary that Y is true" is **false**.

(Verified as a logical derivation; only chaining and contraposition were used.) The audited 2026 Set-A paper contains a four-statement item of exactly this shape.

- **MEMORY: Escalation rule:** two arrows plus one contradiction beat any truth table under exam time.

4. Time-saving techniques (safe conditions)

- ANALYSIS: **Strongest-constraint-first** placement. *Safe always; it minimises backtracking.*
- ANALYSIS: **Counter-diagram test** for syllogisms - try to build one case where the conclusion fails. **Safe always; if you can, the conclusion is invalid.**
- ANALYSIS: **Complement counting** $P(\text{at least one}) = 1 - P(\text{none})$ is always valid. Independence is needed

only when multiplying component probabilities to calculate $P(\text{none})$.

- ANALYSIS: **Inclusion-exclusion** for "either/or" counts. **Safe; remember to subtract the overlap exactly once.**

5. Boundary cases

- ANALYSIS: An arrangement puzzle may have **multiple** valid solutions - then only facts common to **all** solutions are guaranteed.
- ANALYSIS: "Some A are B" does **not** give "some A are not B" - existential statements do not license their negatives.
- ANALYSIS: "It is **not necessary** that Q" is settled by **one** consistent counter-case; it does not mean Q is false.
- ANALYSIS: Probability requires **equally likely** outcomes; weighted or dependent events need the sample space rebuilt.
- ANALYSIS: "Possible" is a **yes/no** verdict, "probability" is a **number** - answer the one asked.

6. Advanced traps

- WRONG: Reporting a placement true in **one** solution when the puzzle has several. -> Report only forced facts.
- WRONG: Accepting a syllogism because it "sounds right". -> Find a counter-diagram.
- WRONG: Reading A -> B as B -> A in a statement-chain item. -> Only contraposition is free.
- WRONG: Applying a coding rule confirmed on only one letter. -> Verify on all given letters.
- WRONG: Multiplying unchanged single-draw fractions for a without-replacement draw. -> Use combinations for an unordered draw, or multiply the changing conditional fractions for a sequence.
- WRONG: Double-counting the overlap in "either/or"; forgetting to add back the triple overlap with three sets. -> Write the full inclusion-exclusion line.
- WRONG: Adding probabilities of non-exclusive events. -> Use inclusion-exclusion or complements.
- WRONG: Reporting a percentage-point move as a percent change in a DI item. -> Name the quantity first.

7. Error analysis

Recurring miss	Master-Framework code	Fix
Over-claiming a unique seat	A	List all solutions; keep only common facts.
Invalid syllogism accepted	C	Always attempt a counter-diagram.
Converse used as if valid	C	Write the arrow and its contrapositive; nothing else.
Wrong coding rule	R	Re-derive the rule on every letter.
Probability sample-space errors	X	Write $6^k/2^k$ or $C(n, r)$ and count favourable explicitly.
DI quantity mismatch	R	Label level / share / change / points before computing.

8. Advanced drill (with full solutions)

ANALYSIS: These items are **new** - they do not repeat Methods 1-4 above or the Foundation file.

1. A fair coin is tossed 3 times. Find $P(\text{at least one head})$.
2. How many diagonals does a regular decagon have?
3. How many distinct arrangements can be made of the letters of "SUCCESS"?
4. How many integers from 1 to 100 are divisible by **at least one** of 2, 3 and 5?
5. Two fair dice are rolled. Find $P(\text{sum} \geq 10)$.
6. A bag holds 4 white and 6 black balls; two are drawn together. Find $P(\text{at least one white})$.
7. Given "every diploma holder is a graduate" and "no graduate is under 18", which of these follow?
(i) No diploma holder is under 18. (ii) Every graduate is a diploma holder.
8. Six people sit around a circle. A is opposite B; C is not adjacent to A. How many distinct arrangements are possible, treating rotations as identical?
9. A bag has 4 white and 6 black balls. Given the first draw was white and it was not replaced, find the probability that the second draw is white.
10. A graph reports output of 60 thousand units and revenue of Rs 120 lakh in Year 1; in Year 2 it reports 75 thousand units and Rs 135 lakh. Find the revenue per thousand units in each year.

Solutions.

1. **7/8**. Complement: $1 - P(\text{no head}) = 1 - (1/2)^3 = 1 - 1/8$. (*Verified.*) The complement is valid here because the tosses are independent.

2. **35**. $n(n-3)/2 = 10 \times 7/2 = 35$. (*Verified.*)

3. **420**. "SUCCESS" has 7 letters with S x3 and C x2: $7!/(3! \times 2!) = 5040/12 = 420$. (*Verified.*)

ANALYSIS: Dividing by only one repeat factorial gives 840 - the standard slip.

4. **74**. Three-set inclusion-exclusion:

$50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$, where the pairwise terms use $\text{lcm}(2, 3) = 6$, $\text{lcm}(2, 5) = 10$, $\text{lcm}(3, 5) = 15$ and the triple term uses $\text{lcm} = 30$. (*Verified by enumeration.*) ANALYSIS: With three sets the triple overlap must be **added back**, not subtracted.

5. **1/6**. Sums of 10, 11 or 12 come from (4,6),(5,5),(6,4),(5,6),(6,5),(6,6) = 6 of 36. (*Verified.*) ANALYSIS: It happens to equal $P(\text{sum} = 7)$ - a coincidence, not a rule.

6. **2/3**. $P(\text{no white}) = C(6, 2)/C(10, 2) = 15/45 = 1/3$, so $P(\text{at least one white}) = 2/3$. (*Verified.*) Draws are **without replacement**, so use combinations, not repeated 6/10.

7. **(i) only**. From diploma $\rightarrow$ graduate and graduate $\rightarrow$ not under 18, chaining gives diploma $\rightarrow$ not under 18, so (i) follows. (ii) is the **converse** of the first premise and does not follow. (*Verified as a logical form.*)

8. **12**. Fix A to remove rotations. B's opposite seat is fixed. C has two permitted remaining seats (not either neighbour of A), and the other three people fill the last three seats in $3!$ ways: $2 \times 3! = 12$. (*Verified.*)

9. **1/3**. After a white first draw, 3 white and 6 black balls remain, so $P(W \mid W) = 3/9 = 1/3$. This is not $4/10$.

10. **2 and 1.8 lakh per thousand units**. Compute $120/60$ and $135/75$; revenue rose while the rate fell. The axes/units must be read before comparing levels.

9. Timed transfer and retry gate

Attempt the ten drills in **15 minutes**. Log a miss as C, A, X, R, or T. A probability retry must show its sample space or conditional fraction; a DI retry must label the axis unit and requested measure. Advance only at **9/10**, with no conditional-probability, overlap-scope, or multi-solution arrangement error; otherwise rebuild the relevant table/diagram and take a fresh timed set.

10. Multi-step data interpretation (worked)

Read this revenue table (in crore):

Year	Product X	All products	X's share
2022	400	1000	40%
2023	500	1250	40%
2024	450	900	50%

- Growth of X, 2022 → 2023 = $(500 - 400)/400 = +25\%$. (Verified.)
- Change in X, 2023 → 2024 = $(450 - 500)/500 = -10\%$. (Verified.)
- ANALYSIS: **Trap 1 - different bases**. A +25% then -10% does **not** return to 400: $400 \times 1.25 \times 0.90 = 450$, matching the table. Percentage changes are on **different bases**, so they do not cancel.
- ANALYSIS: **Trap 2 - level vs share**. From 2023 to 2024 X's revenue **fell** (500 → 450) while X's **share rose** (40% → 50%), because the total fell faster. "X performed worse" and "X gained share" are both true.
- ANALYSIS: **Trap 3 - percentage point vs percent**. That share move is **+10 percentage points** and **+25%** in relative terms (10/40). (Verified.) Expect both numbers among the options.

MEMORY: DI decision rule: before computing, label the quantity asked as **level, share, change in level, change in share (points), or relative change**. Five different numbers live in one table row.

11. Interpersonal and communication scenarios

The official CSAT syllabus separately names **interpersonal skills including communication skills**. The complete Core mechanism now resides in basic/07; optional negotiation and public-dealing depth resides in advanced/07. This section retains the interface with logical option elimination. The provisional 2026 Set-A classification contains six explicit items in this area (Q72-Q77). The 2024/2025 papers do not show a comparable block, so treat this as observed provisional 2026 coverage rather than a fixed annual frequency.

Test	Better response pattern
Stakeholder listening	Identify interests and constraints before proposing action.
Clear communication	Use specific, neutral language; confirm shared understanding.
Conflict handling	De-escalate, separate people from the problem, and seek a fair process.
Ethical decision	Reject deception, coercion, discrimination, and avoidable harm.
Administrative feasibility	Prefer lawful, proportionate, documented action with follow-up.
Relationship/tool matching	Match sender-receiver roles and tools to purpose: inform, consult, persuade, record, or obtain feedback.
Communication barriers	Separate physical/noise, semantic/language, psychological, organisational, and technological barriers.
Questionnaire design	Prefer neutral, specific, single-issue questions with suitable response choices; reject leading or double-barrelled items.

□ Original scenario

A team member repeatedly misses a reporting format because the instructions were ambiguous. The best first response is to clarify the format privately, ask them to restate the requirement, and agree on a check-in. Public blame or silent correction may hide the communication failure rather than resolve it.

□ Original classification drill

1. "Was the training clear and useful?" is **double-barrelled** because it asks two judgements.
2. A technical acronym misunderstood by a citizen is a **semantic barrier**.
3. A post-meeting form used to learn whether instructions were understood is a **feedback tool**.

These drills mirror the provisional 2026 categories without reproducing the source questions.

12. PYQ-pattern notes (2024-2026, Set A)

- ANALYSIS: This is the **second-largest three-year family**. Together with data sufficiency in Topic 05, it accounts for 78 of 240 audited items (32.5%) - see the Master Framework.

- ANALYSIS: Recurring shapes: **seating/ordering arrangements, blood relations and directions, coding-decoding, syllogism/deduction, decision-making scenarios, counting/probability, and data interpretation** from tables/graphs.

- ANALYSIS: The papers reward candidates who **draw the structure** and **read the exact quantity asked**; they punish mental juggling and misread stems.

13. Study links

- FACT: Foundation companion.
- FACT: Algebra, Inequalities and Data Sufficiency - data-sufficiency content overlaps this family.
- FACT: Number Systems and Number Sense - counting and inclusion-exclusion share methods.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-CSAT-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** CSAT
- **Routed question demands:** 127

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	1	3D shape counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	4	Polygon diagonal count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	11	Bar-chart data comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	12	Bar-chart value calculation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	13	Bar-chart ratio reading	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	14	Cube rotation opposite face	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	15	Cube rotation opposite face	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	16	Cube rotation opposite face	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	21	Letter substitution coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	22	Letter-to-number coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	25	Office seating deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	26	Office seating deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	42	Venn-diagram counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	43	Placement logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	44	Placement logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	45	Placement logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	46	Placement statement check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	47	Placement statement check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	48	Placement statement check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	61	Progress graph comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	62	Color arrangement counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	63	Population graph inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	64	Graph statement validity	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	65	Dual-scale graph comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	66	Figural analogy completion	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	67	Graph break-even reading	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	70	Figural rotation series	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	71	Population age-group graph	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	72	Logical relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	73	Policy rate graph inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	74	Table data inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	CSAT	75	Table assumption validity	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2018	CSAT	76	Tax revenue graph inference	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	10	Cube partition cuts	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	13	Syllogism conclusion check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	14	Grid parallelogram count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	20	Set overlap deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	29	Statement sufficiency check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	31	Venn diagram deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	32	Logical chain elimination	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	52	Venn-diagram counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	54	Letter sequence fill	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	56	Denomination combination count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	68	Alphabet substitution coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	70	Sequence conditional count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	CSAT	71	Blood-relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	72	Cube face coloring count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	73	Natural number partition count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	74	Symbol substitution evaluation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	76	Ranking order deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	77	Ranking order deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	78	Ranking order deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	79	Weight inequality deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2019	CSAT	80	Weight inequality deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	15	Letter sequence completion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	19	Blood-relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	32	Syllogism deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	33	Syllogism deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	34	Sequence pair counting	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	CSAT	35	Linear row position count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	36	Age-ordering deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	48	Letter-number coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	49	Letter analogy series	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	52	Table rate comparison	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	60	Denomination subset counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2020	CSAT	67	Constrained permutation count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	10	Syllogism conclusion selection	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	15	Syllogism conclusion selection	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	17	Mirror-image consonant count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	19	Digit-sum integer count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	25	Multi-attribute classification	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	27	Matrix trend missing entry	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	28	Dual-sequence missing entry	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2021	CSAT	35	Venn set overlap range	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	40	Alphabet reversal position coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	45	Table run-rate comparison	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	46	Fractional set deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	49	Multi-conclusion syllogism	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	56	Letter-substitution coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	64	Chessboard diagonal count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	65	Series blank-fill pattern	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	66	Restricted digit arrangement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	69	Non-consecutive selection count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2021	CSAT	74	Pie-chart sector angle	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	10	Digit row arrangement count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	16	Alphabet reversal coding	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	17	Elimination match counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	18	Odd digit counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	24	Letter arrangement counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	27	Route combination counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	29	Letter series position	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	30	Queue minimum size	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	36	Blood relation deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	38	Two-statement syllogism	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	39	Three-statement syllogism	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	40	Circular seating arrangement	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	55	Constrained PIN enumeration	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	56	Circle inscribed triangle count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	70	Pie chart percentage	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	76	Constrained password count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2022	CSAT	77	Grid row arrangement count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	CSAT	79	Form constraint number count	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	4	Worst-case pair selection	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	5	Score partition counting	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	6	Derangement possibility check	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	18	Interior cube count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	19	Position-constrained permutation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	24	Symbol operation decode	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	30	Collinearity-aware triangle count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	36	Worst-case colour group draw	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	37	Letter-shift cipher decode	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	38	Statement-based age ranking	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	39	Family-relation statement sufficiency	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	40	Prime-group odd-one-out	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	48	Circular consecutive selection count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2023	CSAT	54	Circular pass-sequence return	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	55	Sequence middle term	Objective question; official key unavailable locally	Stem verified against official scan; OCR artifact resolved; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	65	Integer-constraint solution count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	66	Adjacent-colour stripe count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	68	Constrained assignment count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	69	Weighing-coin optimisation	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	70	Coded relational chain deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	75	Repeating pattern gap fill	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	77	True-false logic deduction	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	78	Conditional logic conclusion	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	79	Painted cuboid partition count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.
2023	CSAT	80	Reverse alphabetical position count	Objective question; official key unavailable locally	Routed; key unavailable locally	Practise this exact skill form under timed elimination; no answer is inferred here.

What this owner must now support

- 3D shape counting
- Polygon diagonal count
- Bar-chart data comparison
- Bar-chart value calculation
- Bar-chart ratio reading

- Cube rotation opposite face
- Letter substitution coding
- Letter-to-number coding
- Office seating deduction
- Venn-diagram counting
- Placement logic deduction
- Placement statement check
- Progress graph comparison
- Color arrangement counting
- Population graph inference
- Graph statement validity
- Dual-scale graph comparison
- Figural analogy completion
- Graph break-even reading
- Figural rotation series
- Population age-group graph
- Logical relation deduction
- Policy rate graph inference
- Table data inference
- Table assumption validity
- Tax revenue graph inference
- Cube partition cuts
- Syllogism conclusion check
- Grid parallelogram count
- Set overlap deduction
- Statement sufficiency check
- Venn diagram deduction
- Logical chain elimination
- Letter sequence fill
- Denomination combination count
- Alphabet substitution coding
- Sequence conditional count
- Blood-relation deduction
- Cube face coloring count
- Natural number partition count
- Symbol substitution evaluation
- Ranking order deduction
- Weight inequality deduction
- Letter sequence completion
- Syllogism deduction
- Sequence pair counting
- Linear row position count
- Age-ordering deduction
- Letter-number coding

- Letter analogy series
- Table rate comparison
- Denomination subset counting
- Constrained permutation count
- Syllogism conclusion selection
- Mirror-image consonant count
- Digit-sum integer count
- Multi-attribute classification
- Matrix trend missing entry
- Dual-sequence missing entry
- Venn set overlap range
- Alphabet reversal position coding
- Table run-rate comparison
- Fractional set deduction
- Multi-conclusion syllogism
- Letter-substitution coding
- Chessboard diagonal count
- Series blank-fill pattern
- Restricted digit arrangement
- Non-consecutive selection count
- Pie-chart sector angle
- Digit row arrangement count
- Alphabet reversal coding
- Elimination match counting
- Odd digit counting
- Letter arrangement counting
- Route combination counting
- Letter series position
- Queue minimum size
- Blood relation deduction
- Two-statement syllogism
- Three-statement syllogism
- Circular seating arrangement
- Constrained PIN enumeration
- Circle inscribed triangle count
- Pie chart percentage
- Constrained password count
- Grid row arrangement count
- Form constraint number count
- Worst-case pair selection
- Score partition counting
- Derangement possibility check
- Interior cube count
- Position-constrained permutation

- Symbol operation decode
- Collinearity-aware triangle count
- Worst-case colour group draw
- Letter-shift cipher decode
- Statement-based age ranking
- Family-relation statement sufficiency
- Prime-group odd-one-out
- Circular consecutive selection count
- Circular pass-sequence return
- Sequence middle term
- Integer-constraint solution count
- Adjacent-colour stripe count
- Constrained assignment count
- Weighing-coin optimisation
- Coded relational chain deduction
- Repeating pattern gap fill
- True-false logic deduction
- Conditional logic conclusion
- Painted cuboid partition count
- Reverse alphabetical position count

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

Complete revision spine

- **Logic vocabulary:** Separate implication, converse, contrapositive and equivalence.
- **Patterns and series:** Test the simplest stable rule and verify every term.
- **Coding-decoding:** Infer a rule from all examples, not one coincidence.
- **Directions:** Use coordinates and final displacement.
- **Relations:** Build a labelled family graph without assuming gender or generation.
- **Syllogisms:** Use set containment and test existential claims carefully.
- **Arrangements:** Place fixed constraints before flexible ones.
- **Counting:** Apply product, sum, permutation and combination rules with exclusions.
- **Probability:** Define equally likely outcomes and conditional sample spaces.
- **Data interpretation:** Read labels, units, bases and totals before calculating.
- **Decision scenarios:** Separate facts, constraints, objectives and feasible options.
- **Verification:** Enumerate small cases, reverse codes and sanity-check totals.

Ownership and close-option firewall

- **Own here:** Owns series, analogies, coding-decoding, directions, relations, syllogisms, arrangements, ranking, counting, permutations, combinations, probability, tables/charts and scenario logic.
- **Do not duplicate:** Passage inference belongs to Topic 01; algebraic data sufficiency to Topic 05; communication content to Topic 07 even when a matching format is structurally routed here.
- **Verification:** Logical conclusions are model-checked, counting answers are enumerated for small cases, probabilities use explicit sample spaces, and DI preserves denominator and units.

Timed answer route

CLASSIFY -> EXTRACT -> REPRESENT -> EXECUTE -> VERIFY -> DECIDE

- Use estimation or option elimination only after preserving the governing condition.
- A blank costs zero; a rushed unsupported answer also consumes time and may attract negative marks.
- For every error, record concept/application/calculation/reading/passage/time/guess, repair the owner, and retry a new item rather than memorising the old option.